

AARYASH BHAGANKAR

CYBERSECURITY • DIGITAL FORENSICS • SECURITY ENGINEERING • AGENTIC AI

+91 93597 86089 • ceazerdrocks@gmail.com • github.com/automatedcatso • linkedin.com/in/aaryash-bhagankar-50079230a

Cybersecurity-focused engineering student with hands-on cybercrime, digital-forensics and financial-fraud investigation experience at Nashik Cyber Police. Built the NCCRP fund-flow analysis tool that evolved into RED Justice, and now develops security-bounded local AI systems, evidence-linked forensic tooling and applied AI/ML research.

CYBER POLICE EXPERIENCE

Cyber Crime Investigation & Digital Forensics Intern — Cyber Police Station, Police Commissionerate, Nashik City | 17 Jun 2026 - 18 Jul 2026

- Worked across digital forensics, cybercrime investigations, financial-fraud analysis and Legal Department coordination in a live law-enforcement environment.
- Built an internal NCCRP bank fund-flow analysis tool to simplify transaction tracing and day-to-day fraud investigations; later evolved the concept into RED Justice.
- Received a formal Letter of Recommendation from the Senior Police Inspector for technical ability, commitment and professional contribution.

SELECTED SECURITY & AI PROJECTS

RadixOS — Security-Bounded Local AI Command Environment | Electron • Node.js • SQLite/FTS5 • MCP

- Built a local-first Windows agent environment with Hydra evidence-backed project intelligence, multi-provider tool calling, scoped MCP, three-tier execution permissions, guarded shell:false terminal access, secret/path/symlink protections and approval-gated high-risk actions.
- Designed the SQLite/WAL persistence layer with 40 core tables, FTS5 search and a source-provenanced Memory Graph using typed weighted relationships; persists project evidence, tool traces, provider state, jobs, skills, scheduled automations and parallel agent runs.
- Implemented 2-8 lane Agent Mesh workflows with isolated Git worktrees, recovery snapshots, context-compression fidelity receipts, Office automation, public-source retrieval and GitHub-ready CI/CodeQL/dependency-review workflows.

RED Justice — Offline-First Criminal Network & Evidence Intelligence | Flagship • evolution of NCCRP internship tool

- Ingests FIRs, PDFs/DOCX, CDRs, bank statements, chats/emails, screenshots/OCR and structured evidence; deterministic-first extraction validates identifiers, resolves cross-file entities and builds explainable money-flow / relationship graphs.
- Designed a 30-model Prisma/SQLite evidence schema covering immutable evidence versions, transactions, claims/contradictions, graph snapshots, investigator decisions and provenance-backed findings.
- Implemented script-protected key custody, AES-256-GCM encryption, HKDF-separated keys, blind indexes, Ed25519-signed hash chains for custody/decisions and Merkle anchoring for tamper verification.

VisionTrace — Local-First Video Evidence Reconstruction | Digital Forensics • Computer Vision

- Built privacy-focused Browser Lite and loopback FastAPI modes for local video analysis with SHA-256 media hashing, bounded uploads, controlled storage/CORS and evidence-preserving case state.
- Implemented transparent motion detection/tracking, versioned event rules, evidence-linked scene graphs, deterministic case queries and exportable reporting with explicit uncertainty and human-verification safeguards.

AI / ML, RESEARCH & TECHNICAL FOUNDATION

Lowkie — Local AI Model Development Workbench

Local-first workbench for designing, training, fine-tuning and serving compact language models. Built a conversational model designer and real PyTorch pipeline with 10M-20B target recipes, scratch/full/LoRA modes, train-only BPE, masked loss, validation, AdamW, early stopping, hardware preflight and safetensors; includes LOWKIE One (Llama 3.2 1B-class) and a browser classifier lab.

Ongoing Research

AI-Driven Energy Management for Smart Supercapacitor-Powered IoT Networks — adaptive AI-based energy management for resource-constrained IoT systems.

Core Skills

Security: Digital forensics, cybercrime investigation, OSINT, fund-flow analysis, evidence provenance, application/network security fundamentals, CTFs

Engineering: Python, TypeScript/JavaScript, Node.js, Electron, C/C++, SQL, Linux, Git/GitHub, Next.js, FastAPI, Prisma, OpenCV, REST/MCP

AI / Data: Agentic AI, PyTorch, LoRA, BPE/tokenization, local LLMs, safetensors, computer vision, SQLite/WAL, FTS5

EDUCATION & RECOGNITION

B.Tech — SRM Institute of Science and Technology, KTR Campus | Pursuing

BS Degree — IIT Madras, Online Foundation Level | Pursuing

CTFs & Recognition: P3rf3ctr00t 2025 (Team Sn3kBy7e) • ByteBeat 2025 • Yukthi CTF (Chennai Police) • DEFCON CTF 2026 • Nashik Cyber Police Internship Certificate & Letter of Recommendation